

File permissions in Linux

Project description

This task involved taking a Linux directory, establishing the permissions for each file, ensuring that appropriate access was granted to the relevant user and groups and checking the hidden files to ensure that they could not be accessed.

Through the below, I have, using a virtual lab, accessed the system with a bash shell, checked the permissions for the files, and then amended these as required.

Check file and directory details

This document displays the file structure of the /home/researcher2/projects directory and the permissions of the files and subdirectory it contains.

In the /home/researcher2/projects directory, there are five files with the following names and permissions:

- project_k.txt
User = read, write,
Group = read, write
Other = read, write
- project_m.txt
User = read, write
Group = read
Other = none
- project_r.txt
User= read, write
Group = read, write
Other = read
- project_t.txt
User = read, write
Group = read, write
Other = read
- .project_x.txt
User = read, write
Group = write
Other = none

There is also one subdirectory inside the projects directory named drafts. The permissions on drafts are:

User = read, write, execute

Group = execute

Other = none

Describe the permissions string

When displaying the permissions for a file or directory, it is displayed in a 10 character string as below:

drwxrwxrwx - in this d represents that you are displaying the permissions for a directory, if it were a file, this would be replaced with - . Characters 2-4 display the permissions for the user, 5-7 permissions for the group and 8-10 are for others.

R- Indicates read permissions. Able to read the file contents, for directories - able to read all contents in the directory including files and sub directories

W- Indicates write permissions. Able to make modification to file contents, for directories - ability to create new files

X- Indicates execute permissions. Able to execute the files if it's a programme, for directories - the ability to enter the directory and access its files

Change file permissions

In this task, after using the command `ls -l` (this lists the contents of the current directory and then the permissions for each), the system has shown that the `project_k.txt` file has write access for other, which goes against the company policy.

By using `chmod o-w project_k.txt` I amended this permission. In this, `chmod` is the command, `o-w` is the first argument indicating that I am removing write access from other. The second argument is `project_k.txt` which indicated which file I am referring to.

The next file `project_m.txt` should only be read and writable by the user, but currently can be read by group. Again using `chmod g-r project_m.txt`, I remove the access for the group.

Change file permissions on a hidden file

The directory contains one hidden file, `.project_x.txt`. This can be found using the `ls -a` command which lists all files including hidden files. By using `ls -la`, I am able to list all content in the directory including hidden files and their permissions. This has indicated that `.project_x.txt` has the following permissions `rw- -w- - - -`. This file should only be readable by the user and the group.

I amended this by using the following `chmod u=r,g=r .project_x.txt`. By using `u=r` and `g=r` this overwrites the current permissions and sets them to the new indicated permissions.

Change directory permissions

Finally, I checked the drafts directory permissions. At present it was set to `rwX - - X - -`. This directory should only be able to be accessed by the user and therefore required amendment. This was achieved with the following: `chmod g-X drafts`. Removing permissions for the group to execute.

Summary

Above I have examined a company's directories and the access permissions to files. By using commands in the Bash shell, I have amended the files and directory, ensuring that they can be accessed as necessary by the appropriate user/group. `Project_r.txt` and `project_t.txt` were found to be compliant and required no change.